The shortest way

Optimization · Level 1 · Lesson 2



Today we will

- Find a way’s length by adding up its roads.
- Find the shortest way by settling the nearest place first.
- Find a new way when a road closes, and ask who walks the farthest.

Remember

- Looks can fool you: add up the road lengths.
- Settle the waiting place with the smallest number; its number is then final.
- The shortest way depends on today’s map: plan again when a road closes.

Playground Dijkstra

1. Warm-up (5 min): Which way to school is the shortest?
2. Teach (8 min): Add up roads; settle the smallest first.
3. Chalk map (15 min): Pupils settle villages with number cards.
4. Flood a road (7 min): Close one road and plan again.
5. Wrap-up (5 min): Who walks the farthest?

No playground? Lay string between chairs as roads and tape paper labels with the lengths. Keep the same map as the lesson so pupils can check their answers: 13 km, then 14 km.

Let’s discuss

1. Why did the straight-looking road lose?
2. Why can a settled village never get a smaller number?
3. Who walks the farthest to our school, and what would help?

Exit ticket

- Draw a small map with four places and road lengths, and mark the shortest way.